

DSA (Abdul Bari) - Day 2

45-day plan | YouTube playlist course

Time & Space Complexity (Big-O)

Measure how an algorithm scales with input size.

Key concepts

- Big-O describes worst-case growth as n grows.
- Common orders: $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$.
- Count the dominant operations; drop constants.

Key idea / example

Nested loops over $n \rightarrow O(n^2)$; a single loop $\rightarrow O(n)$.

Today's tasks (1.5h)

- ☐ Learn the common orders
- ☐ Find Big-O of 2 loops
- ☐ Rank orders fastest to slowest

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)